MSE 360: Kinetic Processes in Materials
Homework #6

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Equation 4.40 in the text provides the solution for the transient diffusion of a "thin layer" of material between two semi-infinite bodies:
$$C_i(x,t) = \frac{N_i}{\sqrt{4\pi D t}} e^{-x^2/(4Dt)}$$
This thin-film solution yields a Gaussian profile that gradually broadens as a function of time. Derive a mathematical expression quantifying how this Gaussian profile "broadens" as a function of time. Use the peak width at half-maximum (i.e., the distance between the two points where the concentration is one-half of its peak value at any given time) as the definition of peak breadth.

$C_{i,max}$ at $x=0$:

$$C_{i,max}(0,t) = \frac{N_i}{\sqrt{4\pi D t}} \cancel{\exp\left(\frac{-0^2}{4Dt}\right)}^{1}$$

At half-maximum:

$$\frac{C_{i,max}(0,t)}{2} = \underbrace{\frac{N_i}{\sqrt{4\pi D t}}}_{C_{i,max}(0,t)} \exp\left(\frac{-x^2}{4Dt}\right)$$

$$\frac{\cancel{C_{i,max}(0,t)}}{2} = \cancel{C_{i,max}(0,t)} \exp\left(\frac{-x^2}{4Dt}\right)$$

$$\frac{1}{2} = \exp\left(\frac{-x^2}{4Dt}\right)$$

$$x = \sqrt{-4Dt \ln 0.5} \quad \text{at half-maximum}$$

At full-width half-maximum (FWHM):

$$x_{FWHM} = 2x = 2\sqrt{-4Dt \ln 0.5}$$

$$x_{FWHM} = 4\sqrt{-\ln 0.5\, Dt}$$

$$\boxed{x_{FWHM} = 3.33\sqrt{Dt}}$$